

UIPATH AGENTHACK 2026 · MAESTRO BPMN TRACK

Training Fleet & Cloud Resource Manager

An autonomous **cloud-GPU orchestrator** and **humanoid-RL training lifecycle manager** built end-to-end on UiPath Maestro BPMN — agentic decisions, human-approved cost & safety gates.

Agentic AI · Agent Builder

RPA · Cloud provisioning

Action Center HITL

Apache-2.0

THE PROBLEM

Training humanoid robots burns cloud money — silently

Reinforcement-learning skills (walking, grasping) for robots like the Unitree G1 are compute-heavy and expensive. Three failure modes quietly drain budgets:

- ▶ **Idle GPU spend** — H100 / MI300X instances at \$3–5+/hr left running before & after training.
- ▶ **Silent divergence / collapse** — RL runs diverge or flatline for hours before anyone notices.
- ▶ **No audit & provenance** — no record ties a finished checkpoint to its exact GPU cost, metrics, and approvals.

Managing async sim containers, live telemetry, and cloud budgets is a manual, error-prone operational headache.

THE SOLUTION

Governance for ML infra — as a BPMN process

A governed, agentic, human-in-the-loop orchestration on **UiPath Maestro BPMN** that runs the whole training lifecycle: AI agents do the analysis; humans approve the critical cost and physics-safety gates.

Agentic AI

UiPath **Agent Builder** agents classify each training anomaly, score risk, and decide **fast-track vs. human review**.

RPA

A UiPath **RPA Robot** provisions cost-optimal GPUs (AWS/GCP/AMD), SSH-kills bad runs, and writes to cloud consoles with no clean API.

Action Center (HITL)

Researchers approve checkpoints & high-cost actions on rich, read-only AI context.

Active cost containment

Diverging runs are terminated instantly — saving thousands in wasted GPU hours.

HOW IT WORKS

From training request to audited checkpoint

1 · INGEST

Researcher requests a skill (e.g. walk-gait-v2) via a form / GitHub.

2 · BROKER

RPA finds the cheapest available GPU across AWS / GCP / AMD.

3 · DEPLOY

SSH-launches the MuJoCo RL container on the chosen instance.

4 · MONITOR

AI agent watches reward/loss; on collapse → instant SSH shutdown + alert.

5 · HITL

Healthy run → Action Center card; researcher approves the checkpoint.

6 · AUDIT

Register weights, tear down GPU, log immutable cost & approval trail.

An **agentic Fast-Track gateway** auto-resolves low-risk anomalies (e.g. release an idle instance); high-risk / production-bound ones escalate to a human.

ARCHITECTURE

UiPath-core, vendor-agnostic orchestration

3 processes · 8 BPMN containers · 9 swimlane roles. Cloud providers, telemetry sources, LLMs and registries are swappable task nodes invoked by the BPMN flow.

Anomaly taxonomy

GRADIENT_COLLAPSE · LOSS_DIVERGENCE ·
HARDWARE_FAULT · RESOURCE_RISK

Risk → routing

severity + detector confidence × criticality zone (high-cost-gpu
/ production-bound / long-running) → HITL or Fast-Track

Agents

Telemetry Analyst · Anomaly Risk · Cost & Impact ·
Compliance-summary

Execution

RPA → cloud console · billing dashboard · model registry ·
SSH container control

Built on the full UiPath stack

- ▶ **UiPath Maestro (BPMN 2.0)** — the orchestration core (3 processes / 8 containers / 9 roles).
- ▶ **UiPath Agent Builder** — Telemetry Analyst, Anomaly Risk, Cost & Impact, compliance-summary agents.
- ▶ **UiPath Action Center** — human-in-the-loop Researcher Review Board (registered action type + forms).
- ▶ **UiPath RPA Robot + API Workflows** — GPU provisioning & cloud/registry write-back (QueueProcessor.xaml).
- ▶ **UiPath Orchestrator** — Queues (IncidentReports), Folders, Jobs, transaction status.

Agent Type — both

Low-code: UiPath Agent Builder agents orchestrated by Maestro. **Coded:** the same contracts implemented as deterministic, testable Python task nodes (Claude primary / Gemini fallback).

LIVE DEMO

Agentic triage, running end-to-end

8 training-run anomaly samples flow through the pipeline → analyst → risk gateway → action → immutable audit. Live on UiPath Orchestrator via the IncidentReports queue.

8

anomaly events

6

→ HITL review board

2

→ autonomous fast-track

100%

audited & tested

- ▶ **gradient_collapse / loss_divergence / gpu_oom** → high risk → escalated to researcher.
- ▶ **idle_spend** → low risk → auto-resolved (instance released) with zero human touch.

IMPACT

Enterprise control over high-cost ML infrastructure

Save thousands

Active cost-containment loop terminates bad runs instantly and kills idle spend — up to ~60% of wasted cloud GPU hours.

Trustworthy models

HITL verifies AI-trained policies against physics constraints before they reach a physical robot.

Audit & provenance

Every checkpoint is tied to its GPU cost, metrics, and human approval — immutable.

Generic & open

Fully public, Apache-2.0, vendor-agnostic blueprint — usable by any robotics lab.

Roadmap

- ▶ **Decentralized compute** — extend the GPU broker to Vast.ai / Akash for cheaper training.
- ▶ **Multi-agent co-training** — fleets of humanoids co-training in parallel sim instances (MARL).
- ▶ **Multi-framework orchestration** — generalize beyond robotics to any compute-heavy training workload.
- ▶ **Deeper UiPath** — full Maestro Cloud deployment + Action Center reward-curve visualizations.

THANK YOU

Let's make humanoid training governed, cheap & auditable.

github.com/ForenlyAI/uipath-maestro-ecm

Apache-2.0

Claude (primary) · Gemini (fallback)